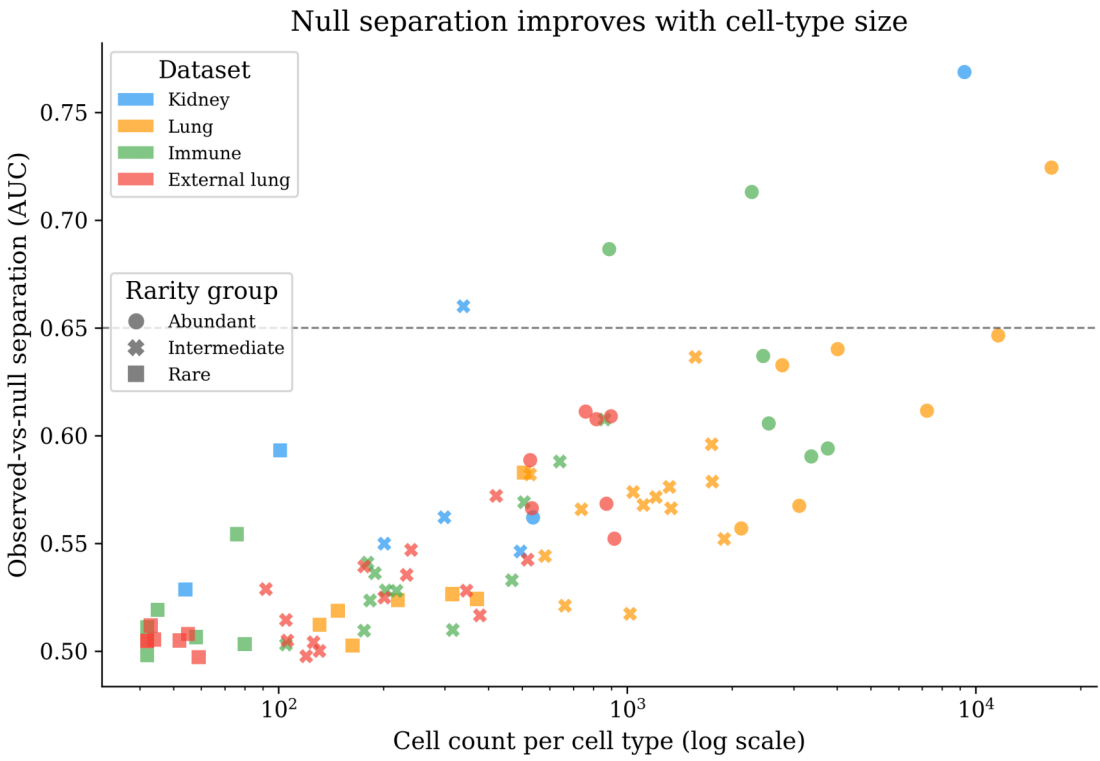
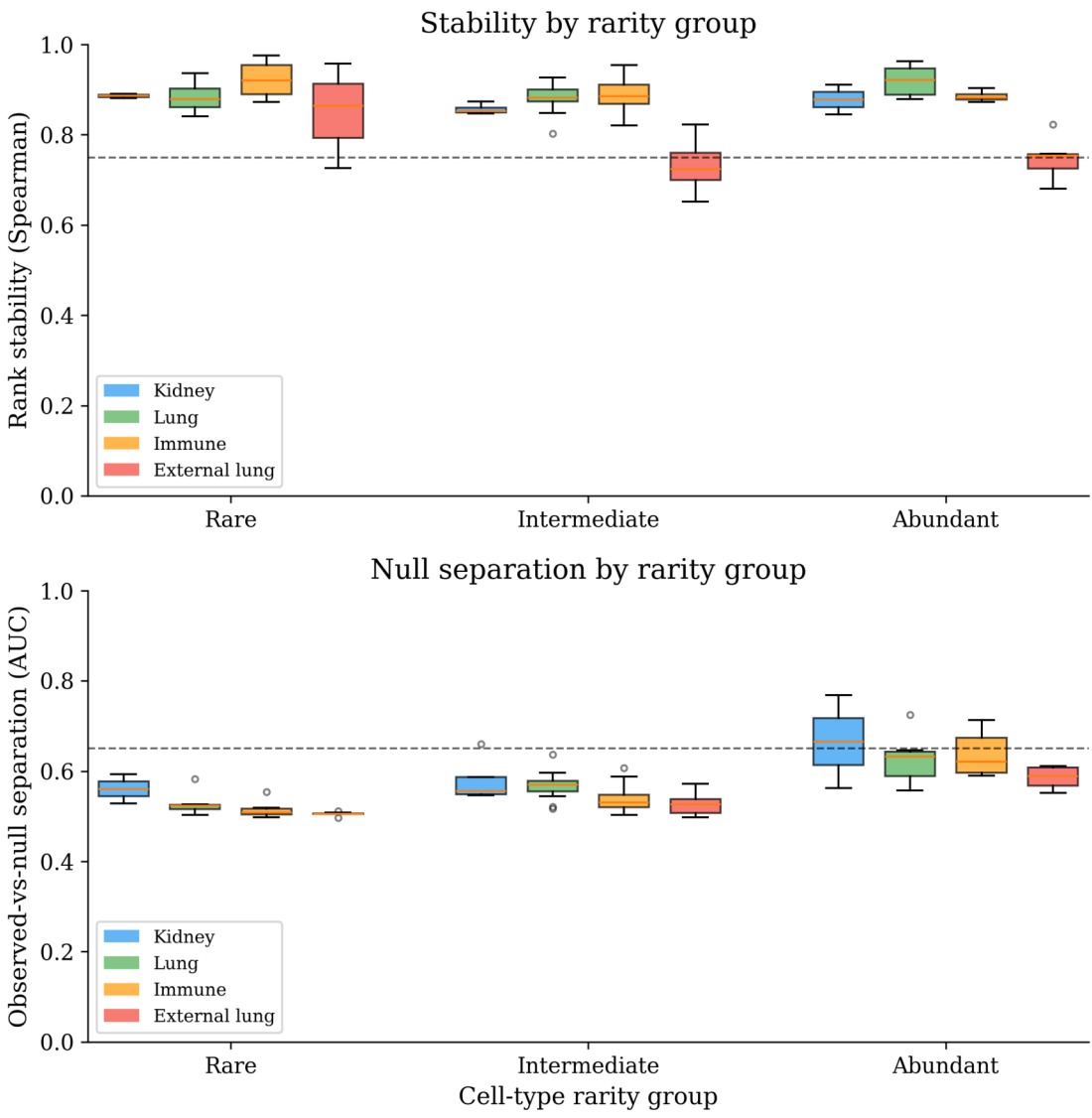
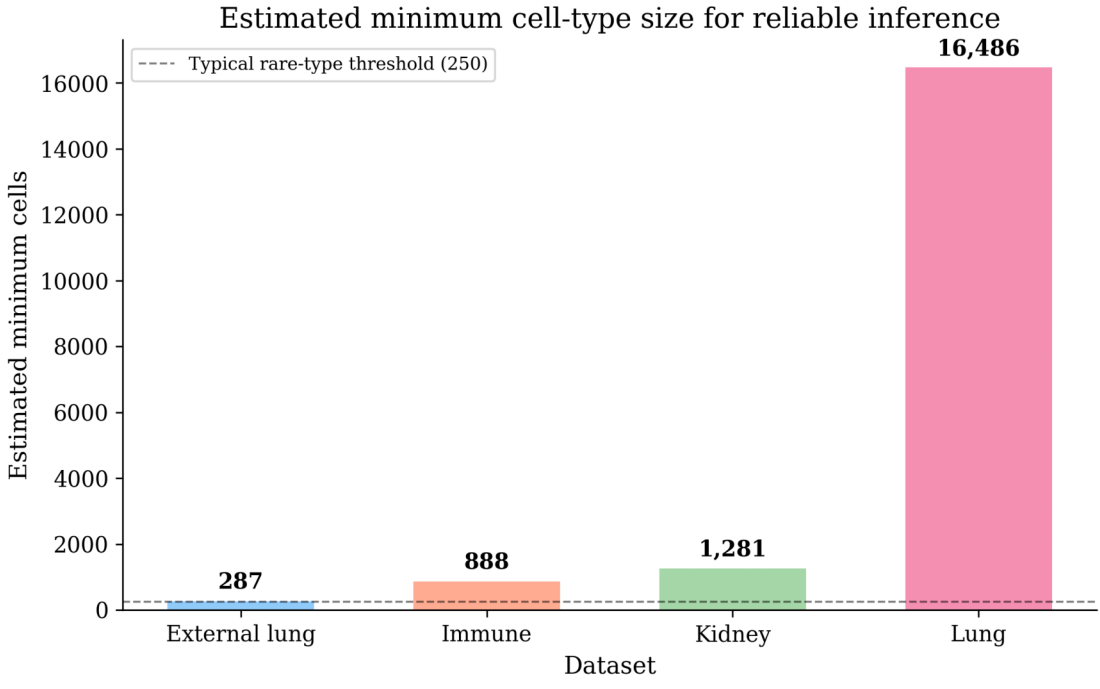
A


Null separation improves with cell-type size
Dataset
Kidney
Lung
Immune
External lung
Rarity group
Abundant
Intermediate
Rare
Observed-vs-null separation (AUC)
0.75
0.70
0.65
0.60
0.55
0.50
10^2
10^3
10^4
Cell count per cell type (log scale)

B


Stability by rarity group
Rank stability (Spearman)
1.0
0.8
0.6
0.4
0.2
0.0
Kidney
Lung
Immune
External lung
Rare
Intermediate
Abundant
Null separation by rarity group
Observed-vs-null separation (AUC)
1.0
0.8
0.6
0.4
0.2
0.0
Kidney
Lung
Immune
External lung
Rare
Intermediate
Abundant
Cell-type rarity group

C


Estimated minimum cell-type size for reliable inference
Typical rare-type threshold (250)
Estimated minimum cells
16000
14000
12000
10000
8000
6000
4000
2000
0
287
888
1,281
16,486
External lung
Immune
Kidney
Lung
Dataset